

ROGER BOS

Weston, Connecticut | (203) - 615 - 3168 | RJBos2@gmail.com | [RogerBos.com](https://www.linkedin.com/in/roger-bos-439060274/)
<https://www.linkedin.com/in/roger-bos-439060274/>

EDUCATION

Norwich University, Northfield, Vermont
Bachelor of Science in Mechanical Engineering
Minor: Mathematics

Graduated: May 2026
GPA 3.60

Honors and Awards

- **Eagle Scout & Quarter Master** – BSA & Sea Scouts (2008 – 2023)
- Dean's List – (Fall 2022 – Spring 2023)
- Presidential Scholarship & Deans Scholarship
- Future Leadership Camp Scholarship
- Take the Lead Scholarship
- United States Naval Academy (USNA) Summer Seminar Graduate

ENGINEERING & RESEARCH EXPERIENCE

Related Work Experience

Modular Optical Mount for Cryostat Integration — Capstone Project, Northfield, Vermont 8/25 – 5/26

- Designed & built a modular cryostat mount for spectroscopy integration at the U.S. Army Research Laboratory
- Built and aligned optical spectroscopy setups to perform temperature-dependent measurements
- Developed 3-axis positioning system using T-slot framing, linear actuators, and ball-screw mechanisms
- Led fabrication and assembly, including machining, drilling/tapping, actuator integration, and system alignment
- Implemented locking casters and leveling solution to improve mobility, repeatability, and lab safety

U.S. Army Research Laboratory — SCIP Internship, Adelphi, Maryland Summer 2025

- Conducted experimental spectroscopy studies on dysprosium-doped samples for visible - IR laser applications
- Assisted in optical characterization including transmission, absorption, and emission measurements
- Worked with laser systems, spectrometers, and precision optical setups
- Analyzed experimental datasets using MATLAB to evaluate emission lifetimes and optical efficiency
- Presented research findings at **SPIE Photonics West** conference

Glasgow Caledonian University — Microgravity Research, Glasgow, Scotland Summer 2024

- Supported experimental hardware preparation and testing for reduced-gravity research environments
- Assisted in mechanical setup, data acquisition, and system verification procedures
- Collaborated in an international research team environment
- Presented research at HMS Dalriada Royal Navy Base in front of British Royal Navy Officers

Carlucci Welding & Fabrication — Drafting & Fabrication, Westport, Connecticut Summer 2025

- Produced drawings used for fabrication via CAD software to support shop manufacturing operations
- Assisted in metal fabrication processes including cutting, fitting, drilling, CNC and assembling of components
- Interpreted engineering drawings and translated design requirements into manufacturable parts
- Collaborated with fabricators to refine designs for manufacturability, tolerances, and structural integrity
- Performed measurements, layout work, and quality checks to ensure dimensional accuracy and fabrication standards

Published Research Paper

SPIE Published Research Paper 13892-37 — SPIE-Intl Soc Optical Eng

Spring 2026

<https://www.spiedigitallibrary.org/conference-proceedings-of-spie/13892/138920N/Comparative-spectroscopic-study-of-Dy3-doped-crystals-and-glasses-for/10.1117/12.3084124.full>

Additional Work Experience

Berman News Service — Head Intern, Wilton, CT

Summer 2023

Weston High School — Robotics Internship, Weston, CT

Spring 2022

Sail Away Sailing School — Sailing Instructor & Marine Maintenance, Bridgeport, CT

Summer 2022

Camp Sequassen — Merit Badge Counselor, New Hartford, CT

Summer 2021

TECHNICAL SKILLS

Engineering & Design

- SolidWorks (CAD & assemblies)
- Fusion 360 / AutoCAD
- Mechanical drafting & fabrication
- Engineering documentation
- Design validation & analysis

Optics & Experimental Systems

- Optical spectroscopy fundamentals
- Laser and optical instrumentation exposure
- Experimental setup & data acquisition
- MATLAB data analysis
- Microsoft Excel (analysis & modeling)

ACTIVITIES & LEADERSHIP

Norwich University Corps of Cadets (2022-2026), Robotics Club Member/President (2016-2024), Brazilian Jiu-Jitsu Club Member/President (2022-2026), American Society of Civil Engineering (2022-2024), Varsity Cross Country Team (2022-2024), Volunteer work at Norfield Congregational Church (2016-2022)

SKILLS

-
- Senior Military College Cadet and student athlete with experience leading teams of 3 to 50 people
 - Experience in C++, Microsoft Office Suite, MATLAB, SolidWorks, and Fusion 360 AutoCAD
 - Possess excellent time management, detail oriented, communication, organization and leadership skills through military training and leadership in the Corps of Cadets, varsity sports, and work experience.